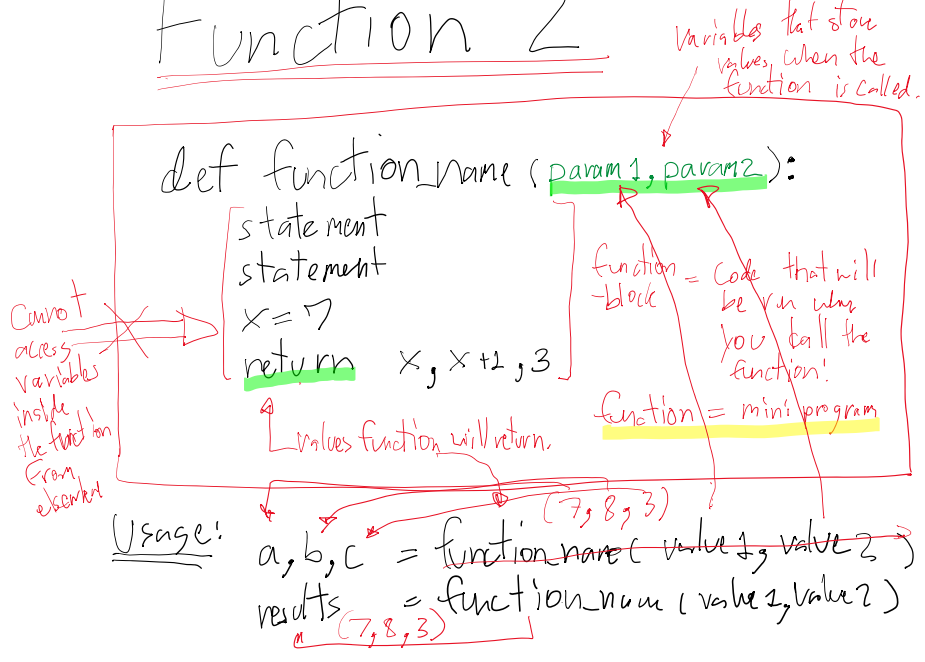
Function 2



How to think about function

1. Function will do one thing — mini program / sub question
   in its own world
2. Function can take in values. ← input into the function
   ↳ Not the same as input from the user!
   store in params
3. Function can return values ← output of the function
   ↳ Not the same as print → output to the screen

Ex1 function to print my name
1) No param
2) No return

```
def print_name():
    print("Nut")
```

Ex2 function to print a name when you call the function
1) one param
2) No return

store a string

```
def print_name(name):
    print("Hello", name)

print_name("Alice")
```

Usage:

Ex3 function to add two things and then print out
1) 2 params

store a number

```
def print_add(x, y)
    print(x+y)
```

and then print out
1) 2 params
2) No return

```
print(x+y)
print_add(2,3)
```

Ex: add two things and return the result
1) 2 params
2) 1 return

```
def add(a,b)
    return a+b → 5
x = add(2,3) 5
print(x) → 5
```

Ex: BMI and obese or not (>30)
1) 2 params: w,h
2) return 2 values BMI, "obese"

main() function
- no param
- no return
- get input from the user
- call BMI_cal(w,h)
- print out the result

```
def BMI_cal(weight, height):
    bmi = weight/height**2
    if bmi > 30:
        return bmi, "obese"
    else:
        return bmi, "Not obese"

def main():
    w = float(input("Enter weight: "))
    h = float(input("Enter height: "))
    bmi, obse = BMI_cal(w, h)
    print(f"{bmi:.2f} {obese}")

main()
```

local variable

Local Variables

```
x = 0
def do_x():
    x = 5
do_x()
print(x)
```

is different one!

local ← x only in do_x()

→ 0

Global Variables

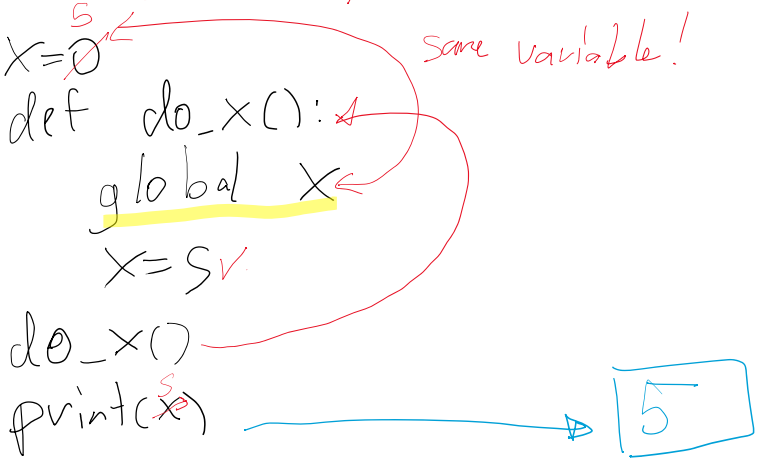
Global Variables

Syntax: global variable_name ← do this inside a function!

↳ make this variable global

↳ meaning that it becomes the same variable that is outside of the function

when you change this variable inside the function outside changes too.



* Named Constant is still here!
  ↳ Sometimes we will call it, Global Constant

Library or Module

↳ Library/Module is a set of functions, typically created by others

↳ You need to import it to use its functions

Syntax: import library_name

Usage: library_name.function_name(...)

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Two libraries: 1) random
2) math

Random Library

- it has a bunch of functions to generate random numbers and do random sampling.

- Function: random.randint(lower, upper)
  - ↳ it returns a random number between [lower, upper] inclusively. (include both.)
    with equal probability.

random.randint(1, 5) —gen. one→ 1 2 3 4 5

prob $\frac{1}{5}$ $\frac{1}{5}$ $\frac{1}{5}$ $\frac{1}{5}$ $\frac{1}{5}$

EX

```
import random
x = random.randint(1,5)
print(x)
```

random → ?

EX8 Data Analysis

1) gen_data(n) fuction → generate a list of random numbers between 1 and 10.
   - ↳ take one value (n)
   - ↳ return a list

↳ return a list

2) mean(numbers) function → cal mean of a given list of numbers
↳ take in a list
↳ return a number

3) main() function → will get n from the user and print out both data and mean
↳ must call other functions.

```
import random
def gen_data(n):
    my_list = []
    for i in range(n):
        x = random.randint(1,10)
        my_list.append(x)
    return my_list
def mean(numbers):          eg. [1,2,3,4]
    return sum(numbers)/len(numbers)
def main():
    n = int(input("Enter n: "))
    data = gen_data(n)
    mu = mean(data)
    print(data)
    print(mu)
main()
```

---

random.seed(int)

random.seed(int)

↳ set the random seed

↳ same random seed will result in same sequence of random numbers.

---

math library

↳ provides math functions and math constants

math functions

math.sin(x)

math.cos(x) ← radius

math.tan(x)

math.log(x)

math.exp(x) → e^x

math.ceil(x) → $\lceil x \rceil$

math.floor(x) → $\lfloor x \rfloor$

math.abs(x) → $|x|$

constants

math.pi → π

math.pi = 3.14...

math.e = 2.71... → e

Summary:

1) Functions
2) global variables
3) Library / Module → random, math